

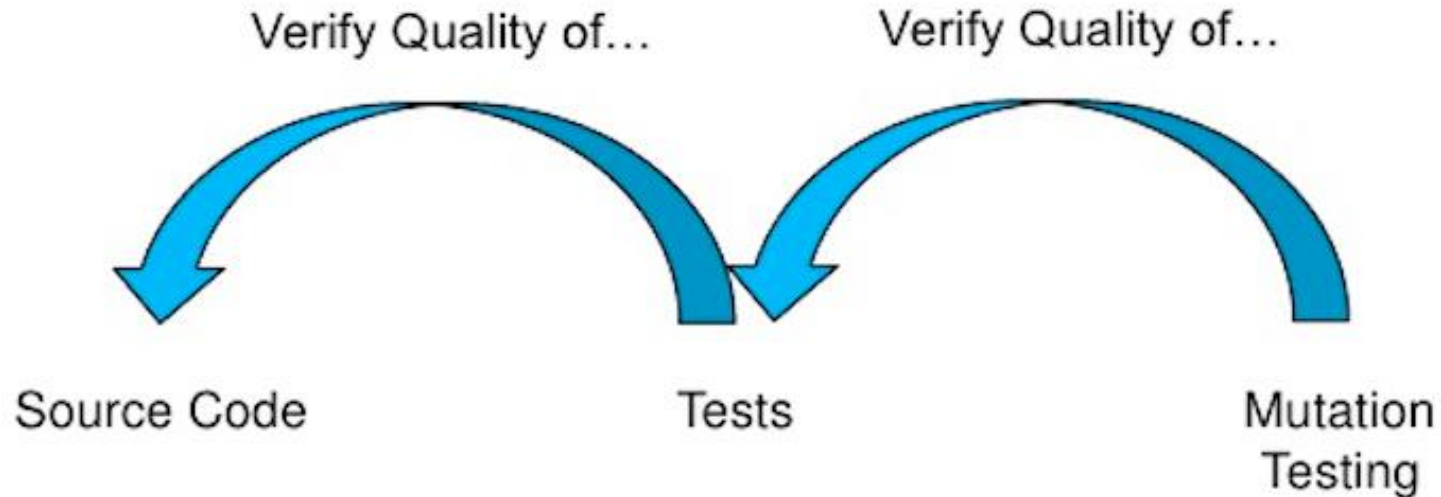
# On Valuable Second-Order Mutants for Mutation Testing

João Paulo Diniz

LabSoft Seminar. Mai 28<sup>th</sup>, 2026

# Introduction

---



# Example of a mutant

---

Original:

```
public class Taxes {  
  
    double simpleTax(double amount) {  
  
        return amount * 0.2;  
    }  
}
```

# Example of a mutant

---

Mutation place:

```
public class Taxes {  
  
    double simpleTax(double amount) {  
  
        return amount * 0.2;  
    }  
}
```

# Example of a mutant

---

\* → +

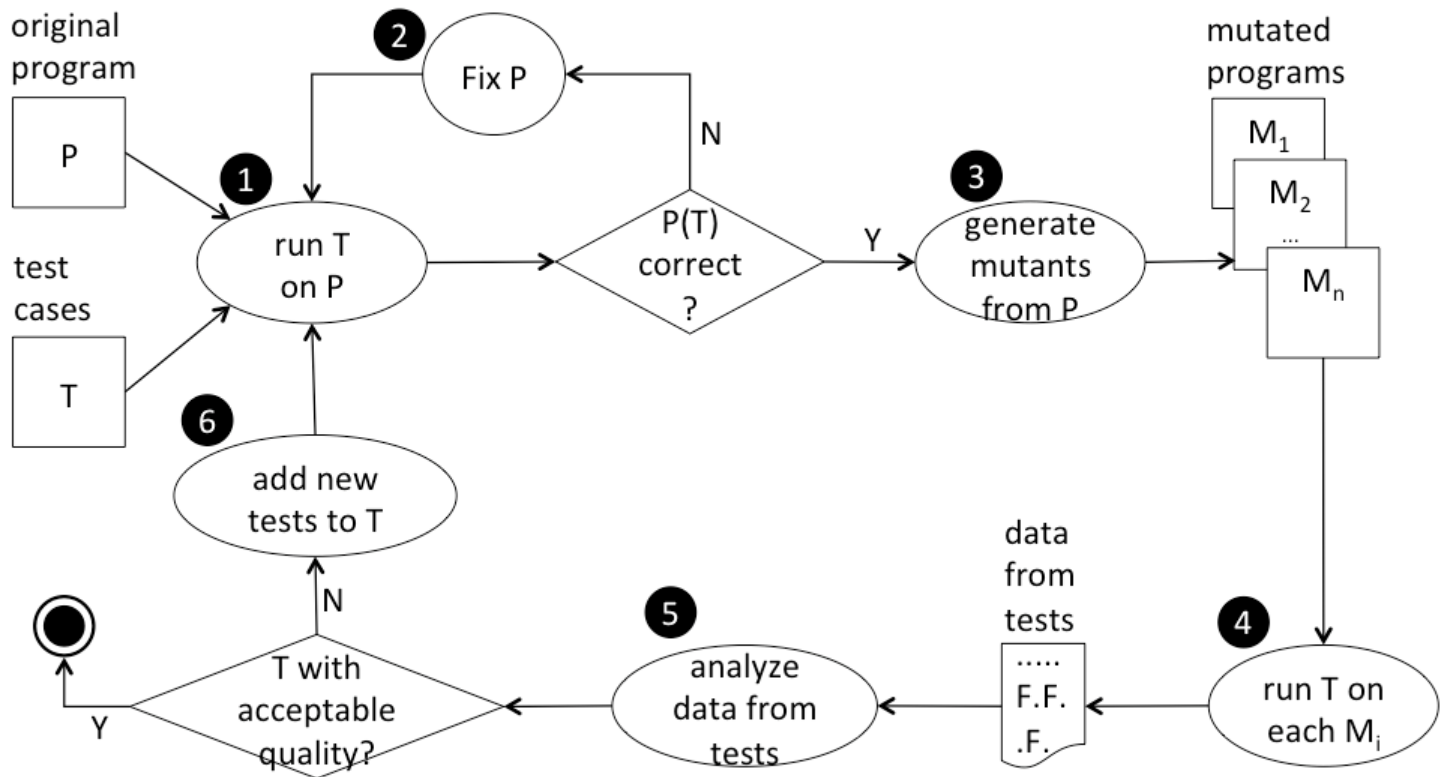
```
public class Taxes {  
  
    double simpleTax(double amount) {  
  
        return amount + 0.2;  
    }  
}
```

# Mutation Testing

---

- ❑ Introducing **artificial changes** into original source code
  - Syntactic changes (**mutations**)
  - Representing real common programming mistakes
  - Changed programs are called **mutants**
- ❑ Running test cases on mutants
  - Result different from original: mutant **killed**
  - Otherwise: **alive**

# Mutation testing process



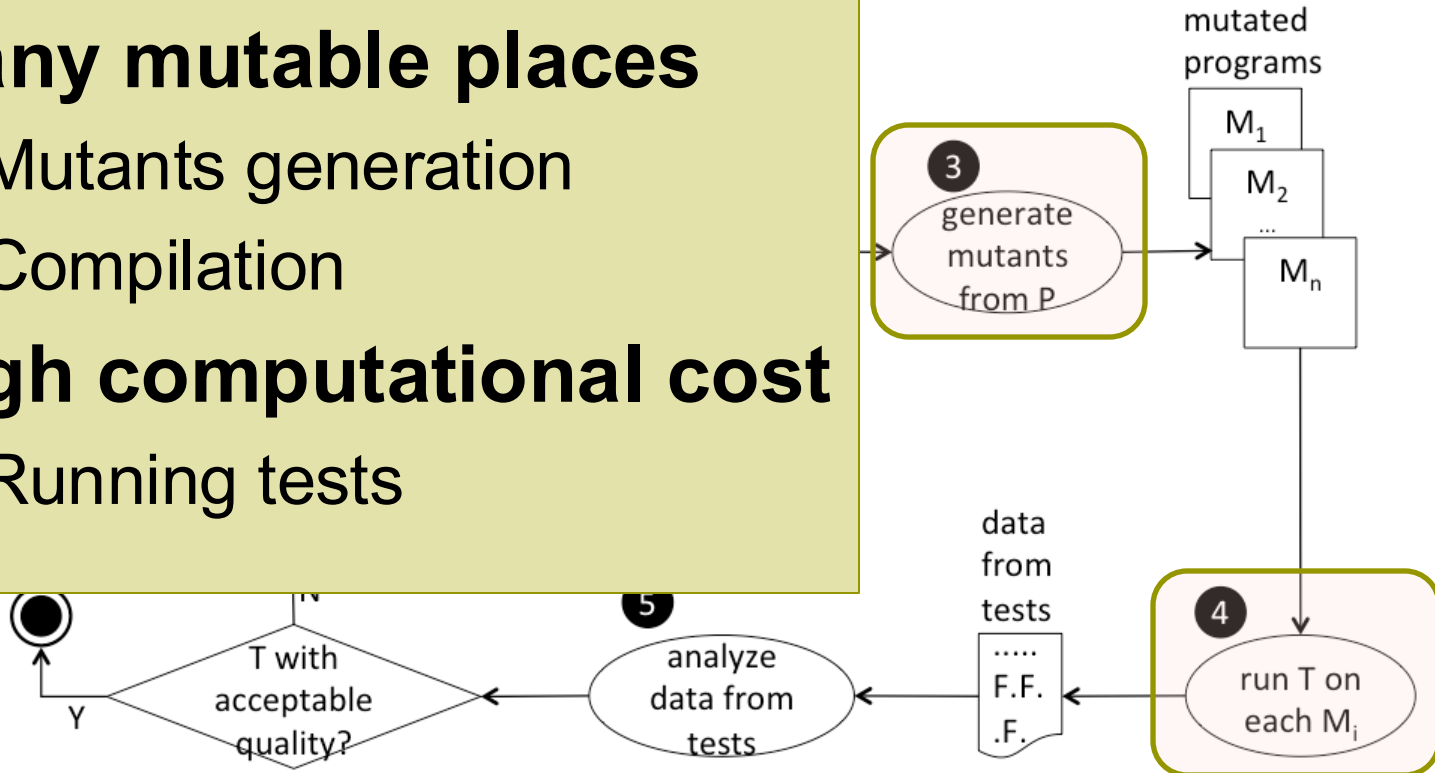
# Mutation testing drawbacks

## 3. Many mutable places

- Mutants generation
- Compilation

## 4. High computational cost

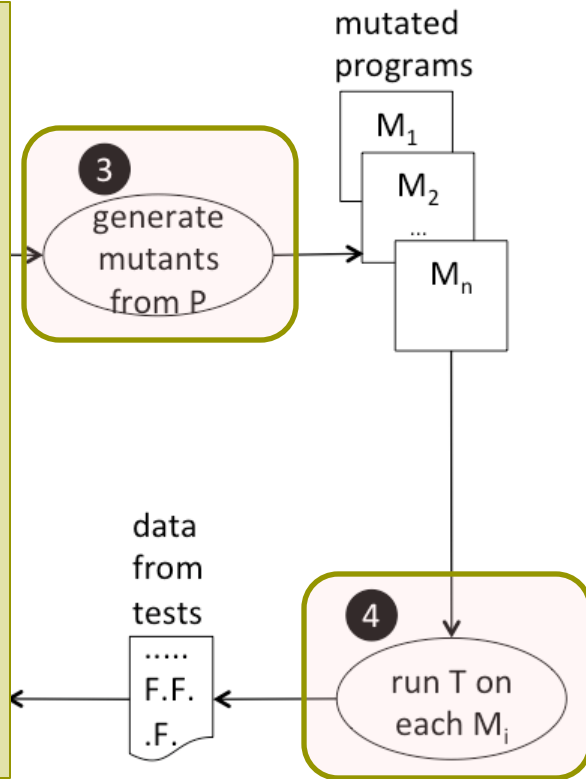
- Running tests



# Mutation testing drawbacks

## ❑ Cost reduction techniques

- Number of mutants
- Number of test cases
- Test case prioritization
- **Higher-order mutation testing**



# Higher-order Mutants

# Higher-order mutants

---

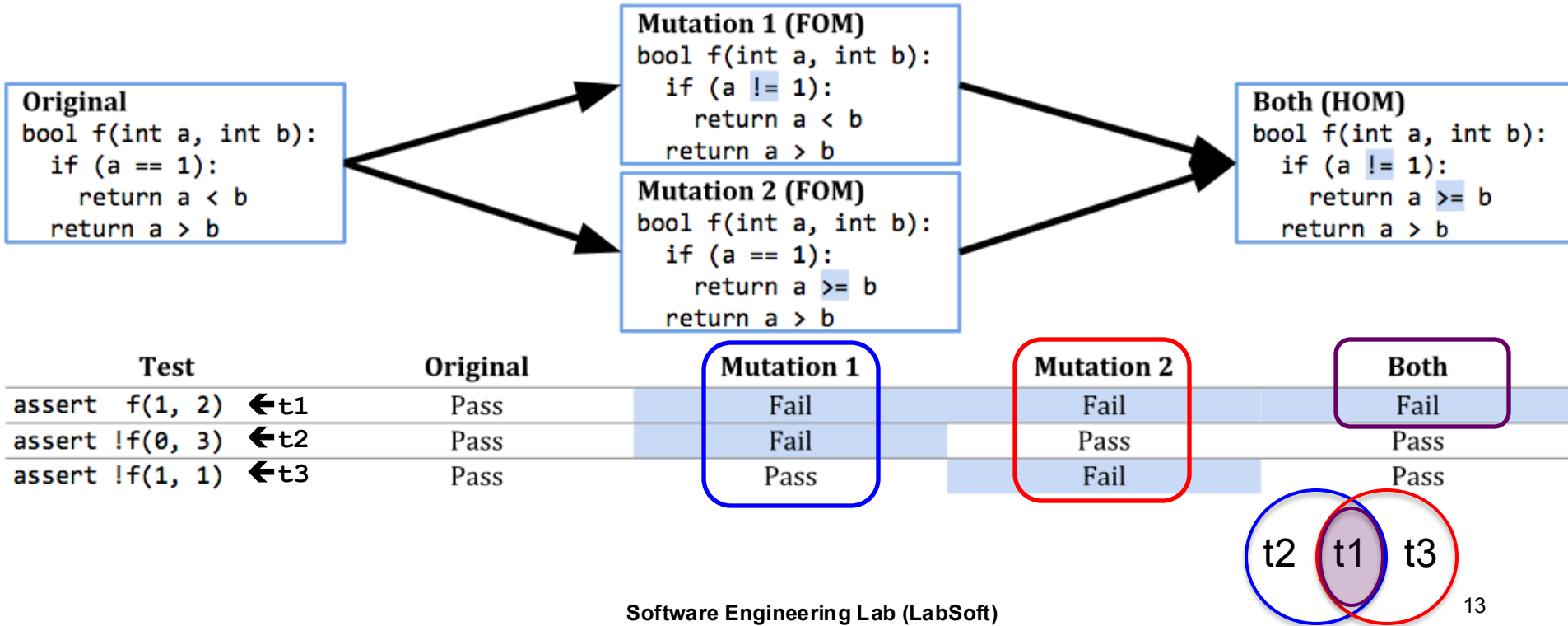
- 1 mutation → first-order mutant (**FOM**)
- > 1 mutations → higher-order mutant (**HOM**)
  - 2 mutations → 2<sup>nd</sup> order mutant (**2OM**)
  
- **n** mutations can generate
  - $\sim C_2^n$  2OMs
  - $\sim C_3^n$  3OMs
  - $\sim 2^n$  HOMs (exponential)

# Higher-order mutants

---

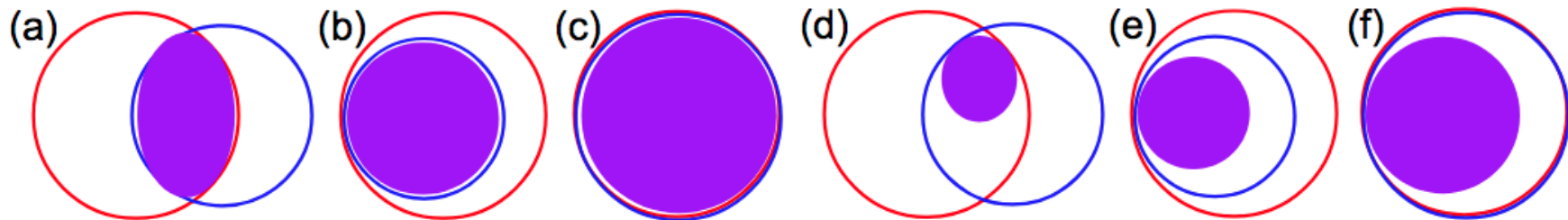
- Most are **more likely to be killed** than FOMs
- But few are **harder** to be killed
  - are **more representative**
  - represent **subtler faults**
  - mutations can (partially) **mask** each other

# A 2OM subsumption example

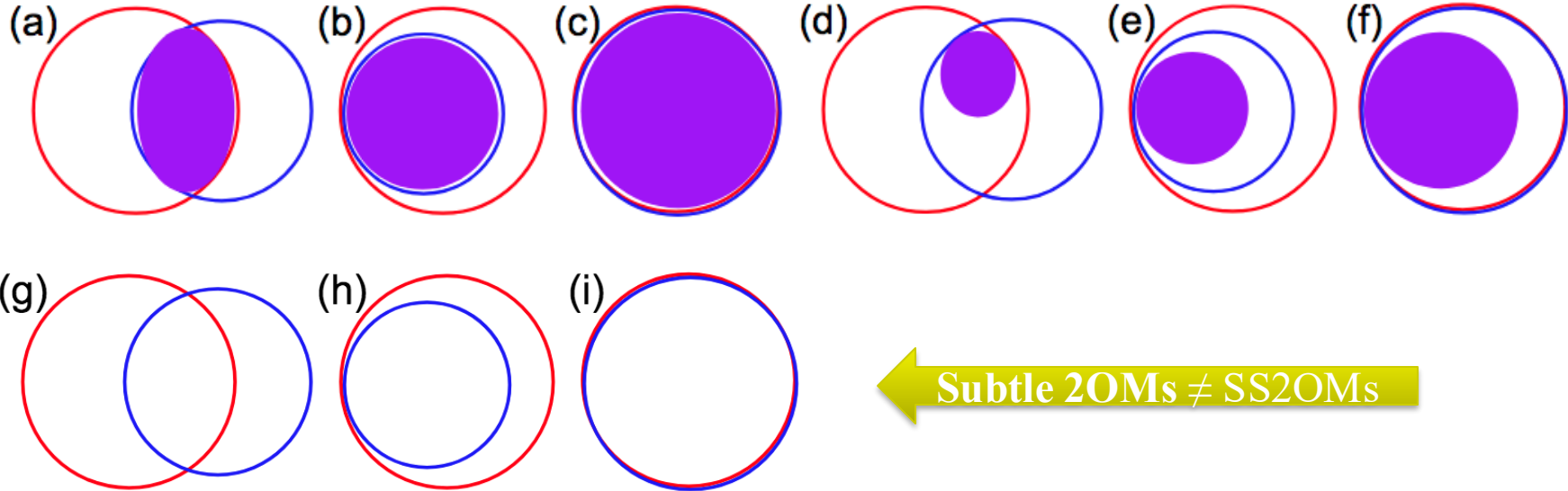
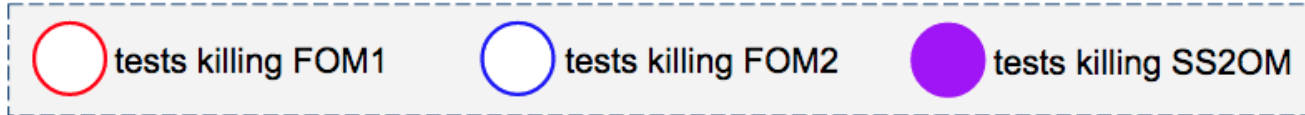


# Venn Diagram “shapes”

---



# Venn Diagram “shapes”



# SSHOMs research

---

- ❑ Exponential search space
- ❑ Dynamic search (running test suite)
- ❑ Trends:
  - Find as many SSHOMs as possible
  - Evolutionary algorithms
    - ❑ Genetic Algorithms
    - ❑ Hill Climbing
  - Multi-objective search

# Research Method

# Main goal

---

- Understand main **characteristics** of SS2OMs
- Retrieve mutants **reduction** via SS2OMs

# Dataset - Java subject systems

| System            | Version    | LOC   | # Tests | JUnit   |
|-------------------|------------|-------|---------|---------|
| Vending Machine   | Exceptions | ~100  | 35      | 4       |
| Triangle          | n/a        | 34    | 12      | 4       |
| Monopoly          | n/a        | 1,181 | 124     | 3       |
| Commons CSV       | 1.8        | ~2k   | 325     | 4       |
| Commons CLI       | 1.4        | 2,699 | 318     | 4       |
| ECal              | 2003.10    | 3,626 | 224     | 3       |
| Commons Validator | 1.6        | 7,409 | 536     | 4       |
| Gson              | 2.9.0      | > 10k | 1,089   | 3 and 4 |
| Chess             | n/a        | 4,924 | 930     | 3 and 4 |
| Joda-Time         | 2.12.7     | ~30k  | 4,232   | 4       |

# Mutation operators implemented

| Operator   | Name                                   | Examples                                                                                   |                                                                                                                   |
|------------|----------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|            |                                        | Original                                                                                   | Mutants                                                                                                           |
| <b>AOR</b> | <b>Arithmetic</b> Operator Replacement | <code>a + b</code>                                                                         | <code>a - b</code> <code>a * b</code><br><code>a / b</code> <code>a % b</code>                                    |
| <b>ROR</b> | <b>Relational</b> Operator Replacement | <code>a &lt; b</code>                                                                      | <code>a &gt; b</code><br><code>a &lt;= b</code> <code>a &gt;= b</code><br><code>a == b</code> <code>a != b</code> |
| <b>LCR</b> | <b>Logical</b> Connector Replacement   | <code>a    b</code>                                                                        | <code>a &amp;&amp; b</code>                                                                                       |
| <b>SBR</b> | <b>Statement</b> Block Removal         | <code>a = b+1;</code><br><br><code>if (...) {</code><br><code>...</code><br><code>}</code> | <code>∅</code><br><br><code>∅</code>                                                                              |



# Results

# FOMs generated

| System            | AOR   | LCR   | ROR    | SBR    | Total  |
|-------------------|-------|-------|--------|--------|--------|
| Vending Machine   | 4     | 1     | 40     | 12     | 57     |
| Triangle          | 36    | 7     | 85     | 10     | 138    |
| Monopoly          | 120   | 8     | 380    | 358    | 866    |
| Commons CSV       | 208   | 56    | 250    | 411    | 925    |
| Commons CLI       | 108   | 80    | 355    | 539    | 1,082  |
| ECal              | 136   | 9     | 315    | 753    | 1,207  |
| Commons Validator | 324   | 174   | 1,675  | 1,024  | 3,197  |
| Gson              | 536   | 211   | 1,430  | 1,535  | 3,712  |
| Chess             | 1,876 | 244   | 1,890  | 1,309  | 5,287  |
| Joda-Time         | 4,300 | 437   | 6,985  | 4,701  | 16,333 |
| Overall           | 7,648 | 1,196 | 13,315 | 10,645 | 32,804 |

# Valuable 2OMs found

| System            | FOMs          | 2OMs               | SS2OMs        | <i>strict-</i> | Pot. Subtle |
|-------------------|---------------|--------------------|---------------|----------------|-------------|
| Vending Machine   | 57            | 1,488              | 25            | -              | -           |
| Triangle          | 138           | 9,086              | 393           | 5              | 13          |
| Monopoly          | 866           | 372,885            | 3,324         | 58             | 46          |
| Commons CSV       | 925           | 425,531            | 4,430         | 15             | 11          |
| Commons CLI       | 1,082         | 582,685            | 1,852         | 41             | 8           |
| ECal              | 1,207         | 727,556            | 1,421         | 26             | -           |
| Commons Validator | 3,197         | 5,102,330          | 17,546        | 121            | 54          |
| Gson              | 3,712         | 6,880,416          | 6,970         | 140            | 79          |
| Chess             | 5,287         | 13,952,357         | 8,959         | 173            | 304         |
| Joda-Time         | 16,333        | 133,394,201        | 36,178        | 374            | 134         |
| <b>Overall</b>    | <b>81,098</b> | <b>161,394,201</b> | <b>81,098</b> | <b>953</b>     | <b>649</b>  |

# Research questions

---

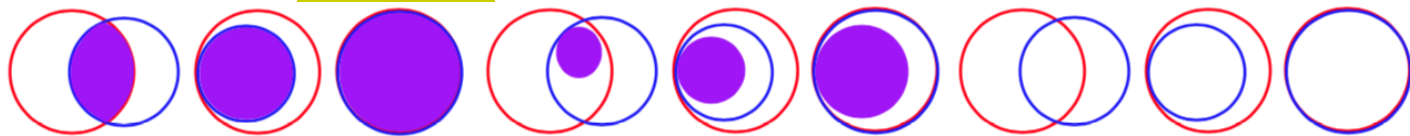
- ❑ **RQ2:** ❑ What is the **distribution** of the mutations in the analyzed Valuable 2OMs?
- ❑ **RQ3:** What are the **most common killing tests** Venn Diagram “shapes” of the Valuable 2OMs found?
- ❑ **RQ6:** What is the **reduction in the number of mutants** that SS2OMs achieve by replacing their constituent FOMs?

# RQ2: Mutations distribution

| System            | SS2OMs        |               |       | strict-SS2OMs |    |           | Pot. Subtle 2OMs |    |    |
|-------------------|---------------|---------------|-------|---------------|----|-----------|------------------|----|----|
|                   | SM            | SC            | DC    | SM            | SC | DC        | SM               | SC | DC |
| Vending Machine   | 20            | 5             | -     | -             | -  | -         | -                | -  | -  |
| Triangle          | 393           | -             | -     | <b>5</b>      | -  | -         | <b>13</b>        | -  | -  |
| Monopoly          | 1,110         | <b>1,470</b>  | 596   | 13            | 3  | <b>42</b> | <b>30</b>        | -  | 16 |
| Commons CSV       | <b>3,422</b>  | 736           | 272   | <b>13</b>     | 2  | -         | <b>11</b>        | -  | -  |
| Commons CLI       | <b>1,627</b>  | 199           | 26    | <b>41</b>     | -  | -         | <b>8</b>         | -  | -  |
| ECal              | <b>761</b>    | 538           | 122   | <b>22</b>     | 4  | -         | -                | -  | -  |
| Commons Validator | 6,416         | <b>10,442</b> | 713   | <b>106</b>    | 15 | -         | <b>54</b>        | -  | -  |
| Gson              | <b>4,291</b>  | 2,541         | 138   | <b>139</b>    | 1  | -         | <b>48</b>        | 31 | -  |
| Chess             | <b>8,738</b>  | 201           | 38    | <b>165</b>    | 8  | -         | <b>304</b>       | -  | -  |
| Joda-Time         | <b>25,932</b> | 7,907         | 2,339 | <b>394</b>    | 25 | -         | <b>118</b>       | 16 | -  |

86.42%

## RQ3: Venn



|                   |     |       |               |   |           |            |            |   |           |
|-------------------|-----|-------|---------------|---|-----------|------------|------------|---|-----------|
| Vending Machine   | -   | 7     | <b>18</b>     | - | -         | -          | -          | - | -         |
| Triangle          | 38  | 143   | <b>207</b>    | - | 4         | 1          | -          | - | <b>13</b> |
| Monopoli          | 1   | 414   | <b>2,851</b>  | 1 | 2         | <b>55</b>  | -          | - | <b>46</b> |
| Commons CSV       | -   | 116   | <b>4,299</b>  | - | 1         | 14         | 3          | - | <b>8</b>  |
| Commons CLI       | 9   | 429   | <b>1,373</b>  | - | 30        | 11         | <b>5</b>   | - | 3         |
| ECal              | 8   | 223   | <b>1,164</b>  | - | 11        | <b>15</b>  | -          | - | -         |
| Commons Validator | 32  | 1,526 | <b>15,867</b> | - | 20        | <b>101</b> | 18         | - | <b>36</b> |
| Gson              | 60  | 1,079 | <b>5,691</b>  | - | <b>73</b> | 67         | 25         | - | <b>54</b> |
| Chess             | 265 | 1,674 | <b>6,847</b>  | 5 | 83        | <b>85</b>  | <b>172</b> | - | 132       |
| Joda-Time         | 31  | 2,322 | <b>33,451</b> | - | 70        | <b>304</b> | <b>21</b>  |   | 113       |

# Mutants reduction - example

---

## FOMs

m1

m2

m3

m4

m5

m6

m7

m8

# Mutants reduction - example

---

**FOMs**

**SS2OMs**

m1

[m1,m2]

m2

[m3,m4]

m3

[m5,m6]

m4

m5

m6

m7

m8

# Mutants reduction - example

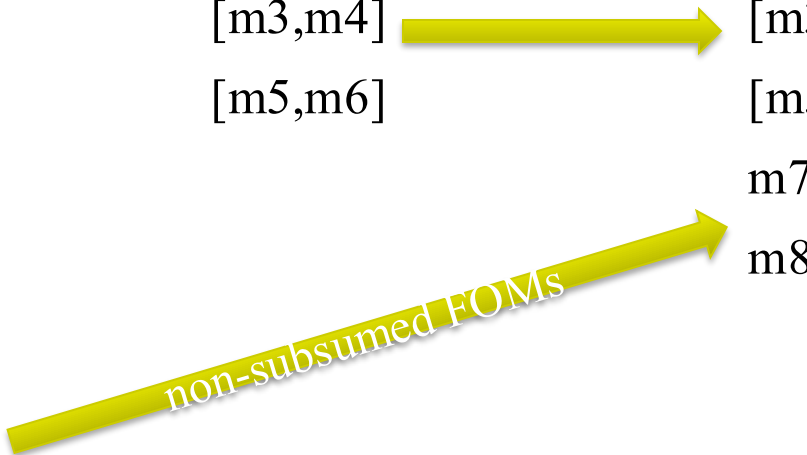
---

| FOMs          | SS2OMs  | Resulting mutants |
|---------------|---------|-------------------|
| <del>m1</del> | [m1,m2] | [m1,m2]           |
| <del>m2</del> | [m3,m4] | [m3,m4]           |
| <del>m3</del> | [m5,m6] | [m5,m6]           |
| <del>m4</del> |         |                   |
| <del>m5</del> |         |                   |
| <del>m6</del> |         |                   |
| m7            |         |                   |
| m8            |         |                   |

# Mutants reduction - example

---

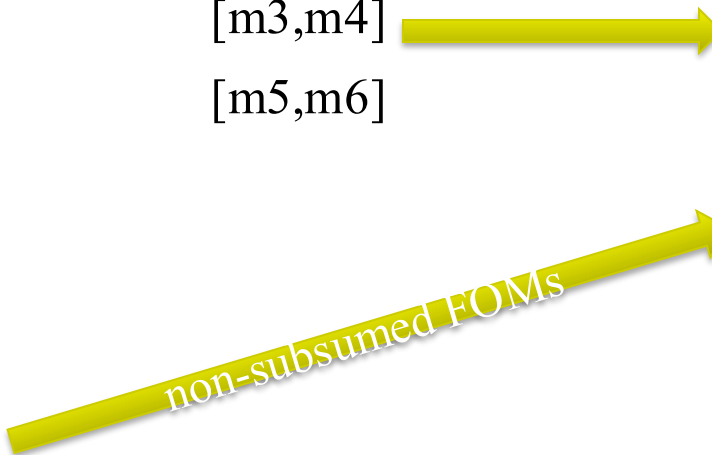
| FOMs          | SS2OMs  | Resulting mutants |
|---------------|---------|-------------------|
| <del>m1</del> | [m1,m2] | [m1,m2]           |
| <del>m2</del> | [m3,m4] | [m3,m4]           |
| <del>m3</del> | [m5,m6] | [m5,m6]           |
| <del>m4</del> |         | m7                |
| <del>m5</del> |         | m8                |
| <del>m6</del> |         |                   |
| m7            |         |                   |
| m8            |         |                   |



non-subsumed FOMs

# Mutants reduction - example

| FOMs          | SS2OMs  | Resulting mutants | Reduction                     |
|---------------|---------|-------------------|-------------------------------|
| <del>m1</del> | [m1,m2] | [m1,m2]           | 3 out of 8 mutants<br>(37.5%) |
| <del>m2</del> | [m3,m4] | [m3,m4]           |                               |
| <del>m3</del> | [m5,m6] | [m5,m6]           |                               |
| <del>m4</del> |         | m7                |                               |
| <del>m5</del> |         | m8                |                               |
| <del>m6</del> |         |                   |                               |
| m7            |         |                   |                               |
| m8            |         |                   |                               |



non-subsumed FOMs

# With much more SS2OMs

---

**FOMs**

**SS2OMs**

**Resulting mutants**

m1 [m1,m2], [m1,m4], ...,

m2 [m2,m3], ..., [m2,m8],

m3 [m3,m4],..., [m3,m8],

m4 [m4,m5], ..., [m4,m8],

m5 [m5,m6], [m5,m7], ...,

m6 [m6,m7], [m6,m8],

m7 [m7,m8], [m1,m7], ...,

m8 [m2,m4], ..., [m5,m8]

??????????

**Minimum  
Set Cover  
Problem**

# MSC by Integer Linear Programming

$$\begin{array}{ll}\text{minimize} & \sum_{j=1}^n x_j \\ \text{subject to} & \sum_{j:v \in S_j} x_j \geq 1 \quad \forall v \in U, \\ & x_j \in \{0, 1\} \quad \forall j \in \{1, \dots, n\}.\end{array}$$

# RQ6: Mutants reduction (1/2)

| System            | SS2OMs        | SS2OMs red.  | SS2OMs discarded | Non-sub. FOMs | Resulting mutants |
|-------------------|---------------|--------------|------------------|---------------|-------------------|
| Vending Machine   | 25            | 17           | 32.00%           | 32            | 49                |
| Triangle          | 393           | 52           | 86.77%           | 36            | 88                |
| Monopoli          | 3,176         | 270          | 91.50%           | 387           | 657               |
| Commons CSV       | 4,430         | 234          | 94.72%           | 503           | 737               |
| Commons CLI       | 1,066         | 311          | 70.83%           | 545           | 856               |
| ECal              | 1,488         | 366          | 75.40%           | 568           | 934               |
| Commons Validator | 17,571        | 944          | 94.63%           | 1,464         | 2,408             |
| Gson              | 6,970         | 976          | 86.00%           | 1,978         | 2,954             |
| Chess             | 8,977         | 1,402        | 84.38%           | 2,832         | 4,234             |
| Joda-Time         | 36,178        | 2,531        | 93.00%           | 11,801        | 14,332            |
| <b>Overall</b>    | <b>81,098</b> | <b>7,161</b> | <b>91.17%</b>    | <b>19,951</b> | <b>27,112</b>     |

# RQ6: Mutants reduction (2/2)

| System            | FOMs   | Resulting mutants | Mutants reduction |
|-------------------|--------|-------------------|-------------------|
| Vending Machine   | 57     | 49                | 14.04%            |
| Triangle          | 138    | 88                | 36.23%            |
| Monopoli          | 866    | 657               | 24.13%            |
| Commons CSV       | 925    | 737               | 20.32%            |
| Commons CLI       | 1,082  | 856               | 31.05%            |
| ECal              | 1,213  | 934               | 22.54%            |
| Commons Validator | 3,197  | 2,408             | 24.52%            |
| Gson              | 3,712  | 2,954             | 20.42%            |
| Chess             | 5,319  | 4,234             | 20.38%            |
| Joda-Time         | 16,333 | 14,332            | 12,25%            |
| Overall           | 32,804 | 27,112            | 17.35%            |

# Threats to validity and limitations

# Threats to validity

---

## ❑ **External:**

- Java
- 10 systems
- 4 MutOps

## ❑ **Construction:** possible errors in our tool

## ❑ **Internal:**

- Only one large system
- Original test suites may have poor quality

# Limitations

---

- ❑ Removal of some
  - Mutants
  - Test cases
  
- ❑ Significant time required to run for the largest systems

# Related work

# 2<sup>nd</sup> order mutation testing

---

- 2OMs sampling
  - Random
  - LastToFirst
  - Different operators
  - etc.
- [Polo et al., 2009], [Kintis et al., 2010], [Madeisky et al., 2013] ...

# SSHOMs

---

- [Jia et al., 2008 and 2009]
  - First search-based algorithms proposed
  - SSHOMs examples
- [Harman et al., 2014]
  - Reduction via greedy minimization
- [Wong et al., 2020]
  - Search for all possible SSHOMs via Variational Execution
  - Search<sub>pri</sub> using some characteristics

# Conclusion and future work

# Conclusion

---

- ❑ Many SS2OMs discarded
- ❑ Mutants reduction via SS2OMs: 17,35%
- ❑ Most mutations in the same class
- ❑ Most SS2OMs and constituent FOMs have the same killing tests

# Future work

---

- ❑ New mutation operators
- ❑ New systems
- ❑ New analyses
- ❑ Investigate how automated source code generation could benefit from mutation testing



# Questions?

